Lecture 2

Cartilage, tendon, ligament materials and biomechanics

OpenSim overview



References:
Biomechanics of Movement, Uchida & Delp, 2020
Biomechanical Basis of Human Movement, Hamill, Knutzen, Derrick, 2014
Basic Biomechanics of the Musculoskeletal System, Nordin and Frankel, 2001

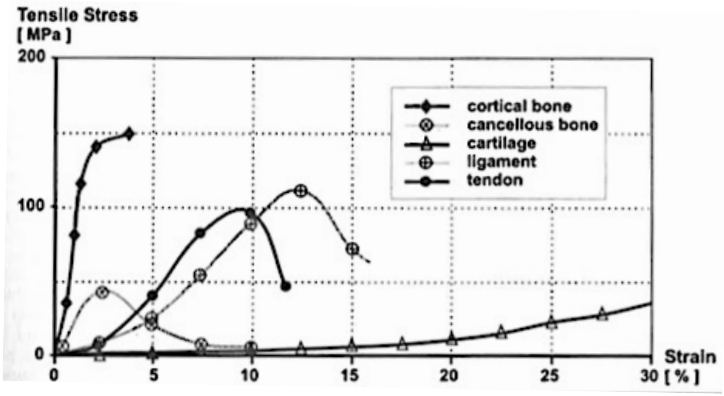
1

Course registration?

Appointments for experiments Thursday available on Canvas

2

Musculoskeletal tissue stress-strain properties



In tensile loading

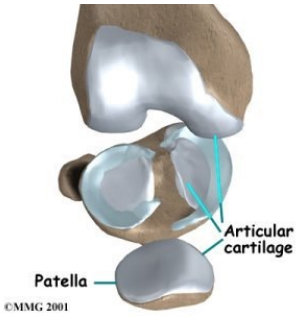
3

Collagen

- Most abundant protein in body
- Inhomogeneously distributed in articular cartilage
- Type I: most abundant. In bones, intervertebral discs, skin, miniscus, tendons, ligaments.
- Type II: Primary type in cartilage
- Stiff and strong in tension
  - Tendons (80% collagen dry weight): tensile stiffness 10^3 MPa. Steel: 220×10^3 MPa
- Weak in compression

4

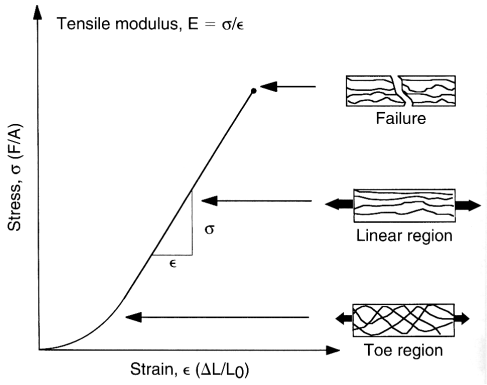
Articular (hyaline) cartilage



- 1-6 mm hyaline articular cartilage in young joints
- Withstands high loads without failure (in most cases)
- Isolated, no blood vessels, lymphatic channels or innervation.
- 60-80% water. Rest is a solid matrix of collagen and proteoglycan, which determine mechanical behavior
- Least cell-dense tissue
- 2 functions in joints:
  - Distribute loads over wide area
  - Allow relative motion with minimal wear and friction
- Anisotropic and viscoelastic
- Coefficient of friction 0.01 to 0.04 (ice at 0° C: 0.10)
- Thickness respond to activity throughout life

5

Articular cartilage strength in tension



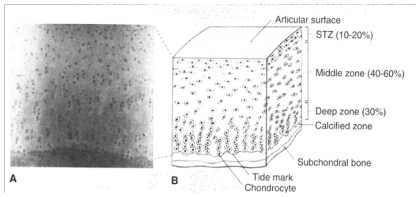
Nordin & Frankel 2001

6

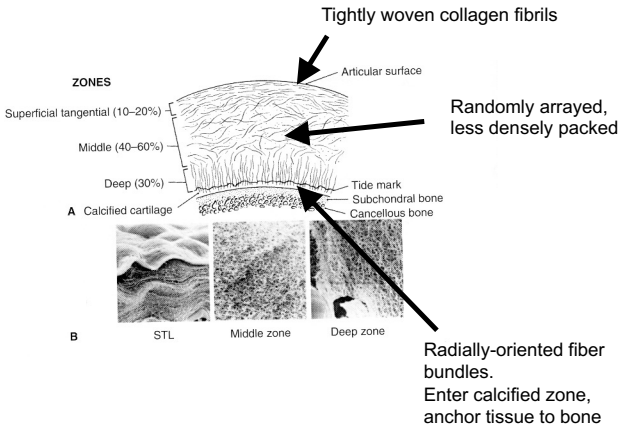
Articular cartilage

Chondrocytes

- Less than 10% volume
- Create and maintain organic elements in extracellular matrix (ECM)



Collagen II

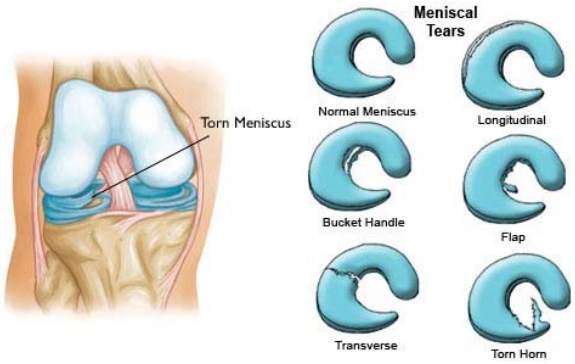


Nordin & Frankel 2001

7

Fibrocartilage

- Often found where articular cartilage meets a tendon or a ligament
- Ex. Knee meniscus
- High concentration of Type I collagen



8

Cartilage wear mechanisms

- Interfacial (interaction of bearing surfaces)
  - No lubricant film between surfaces
  - Either
    - Adhesion (surface fragments adhere and are torn off during sliding)
    - Abrasion (soft material scrapted by harder one)
  - Injury tends to lead to 'downward spiral'
- Fatigue
  - Acculumation of microscopic damage from loading
- 3rd mechanics of damage: impact loading

9

Tendons and Ligaments



- Passive materials, yet essential role in joint motion
- Ligaments: Bone to bone
  - Provide joint stability
  - Guide joint motion
  - Prevent excessive motion
- Tendons: Muscle to bone
  - Transmit tensile load from muscle to bone
    - > produce motion or maintain position
  - Enable muscle belly to be at optimal distance from joint
- Largely composed of collagen type 1
- Few cells (fibroblasts). Fibroblasts produce collagen
- Few blood vessels

10

Patellar Tendon



Parallel collagen fibers

Unmineralized fibrocartilage

Mineralized fibrocartilage

Cortical bone

11

Ligament stress vs. strain

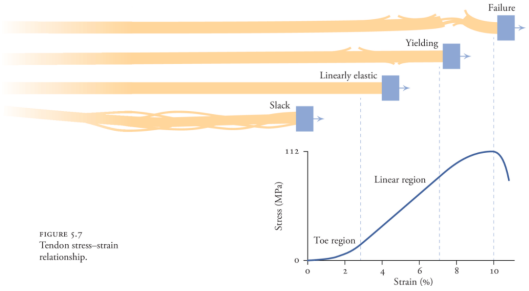


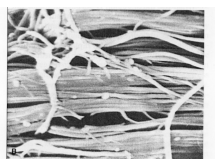
FIGURE 5.7
Tendon stress–strain relationship.

Uchida and Delp 2021

Ex. knee ligament

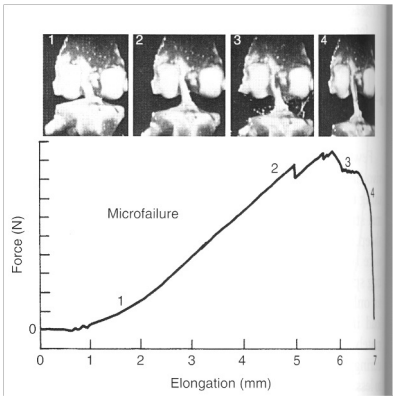


Unloaded



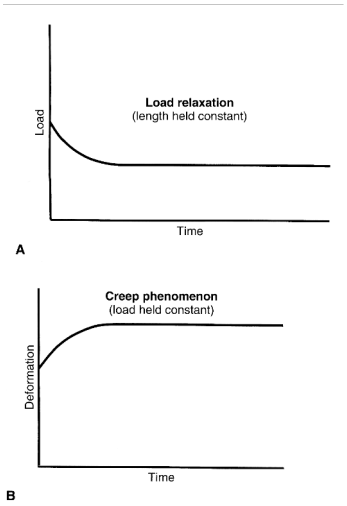
Loaded

Failure: microfailures

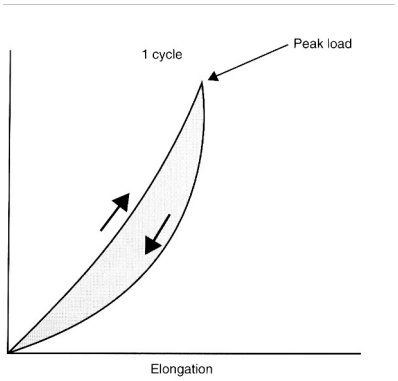


12

Viscoelasticity and hysteresis



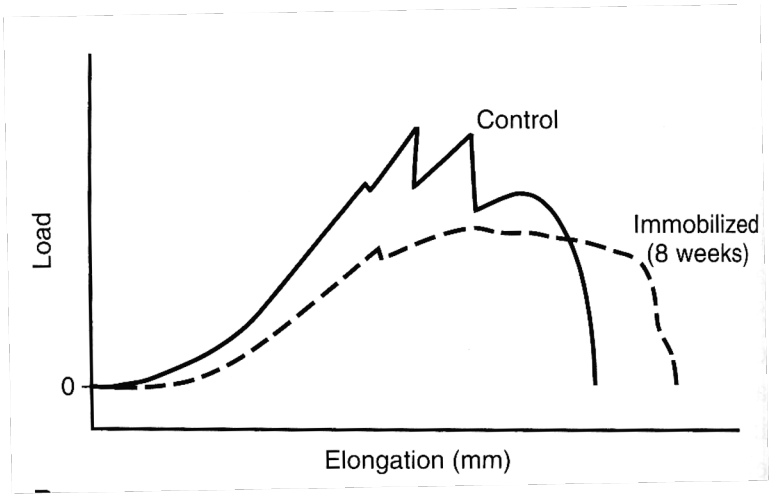
Viscoelastic



Hysteresis

13

Sensitive to underuse



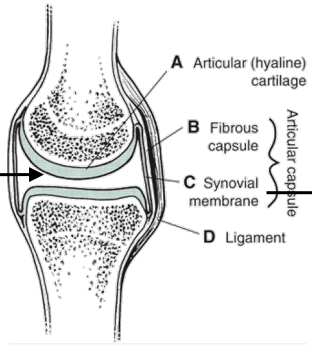
14

Joints/bony articulations

15

Diarthrodial joints

reduced atmospheric pressure

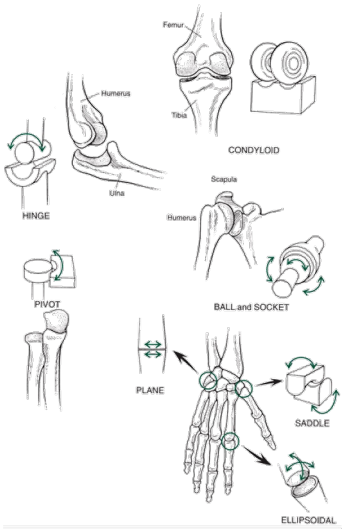


Vascularized, secretes synovial fluid

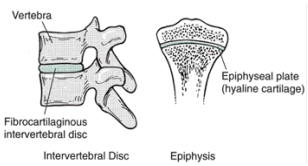
Joint stability is created by:
- Ligaments
- Gravity
- Vacuum

16

7 types of diarthrodial joints

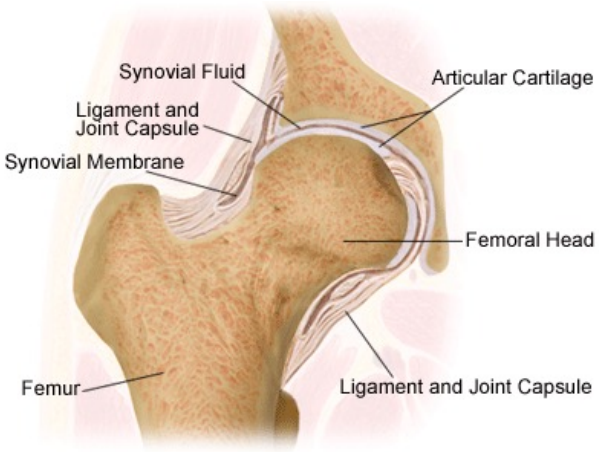


Fibrous joints



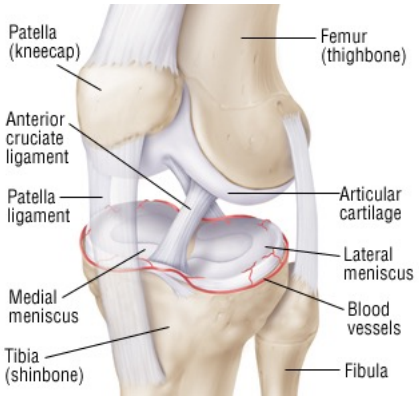
17

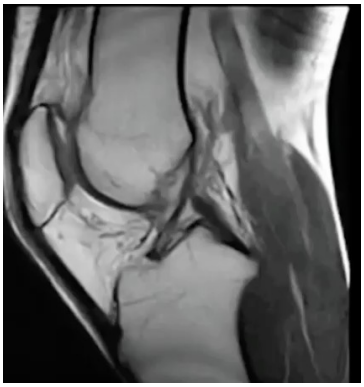
Hip joint



18

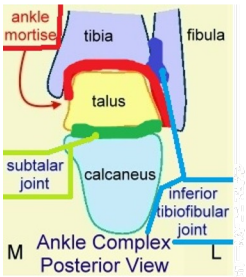
Knee joints

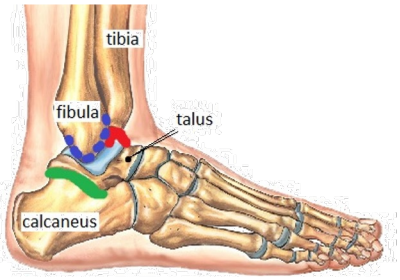




19

Ankle joints

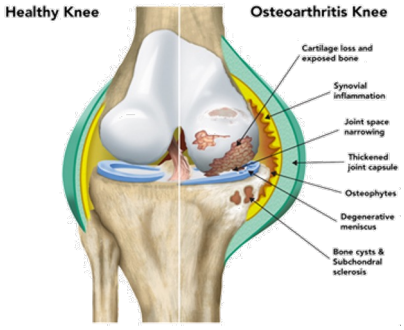




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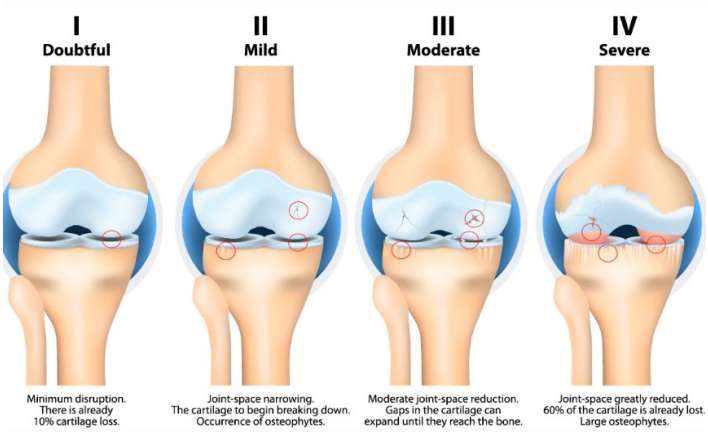
Osteoarthritis

- Primary cause unknown
- Injury of the joint structures, resulting from repeated load
- In articular cartilage, mechanical wear leads to fissures and degeneration
- Leading cause of disability >65 years
- No clear relation to hyperlaxity or osteoporosis or "normal" activity, even strenuous
- Previous injury may accelerate OA
- Increased risk in some occupations and sports/exercises
- Immobilization can be cause
  - cartilage require loading and compression to exchange nutrients and wastes. Some exercise is frequently recommended.



21

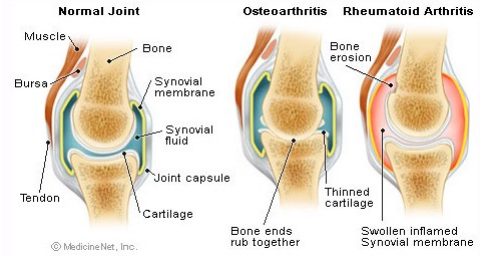
Osteoarthritis



22

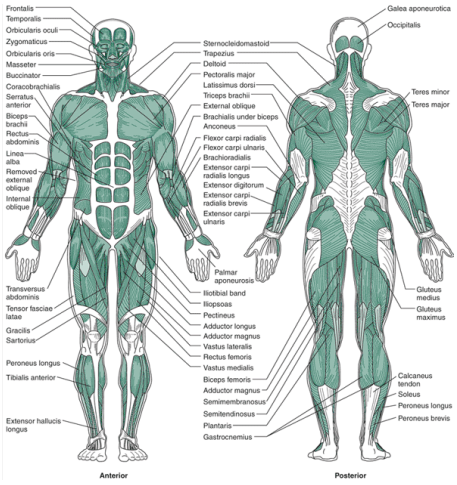
Rheumatoid arthritis

- Auto-immune disease. Precise cause is unknown.
- Can occur in children and adults. Occurs 3x more often in women than in men.
- Synovial membrane attacked
- Joint inflammation and pain



23

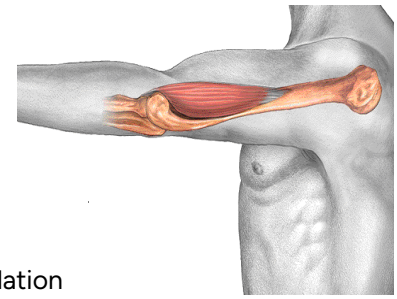
Skeletal Muscle



Hamill & Knutzen 2009

24

Muscle

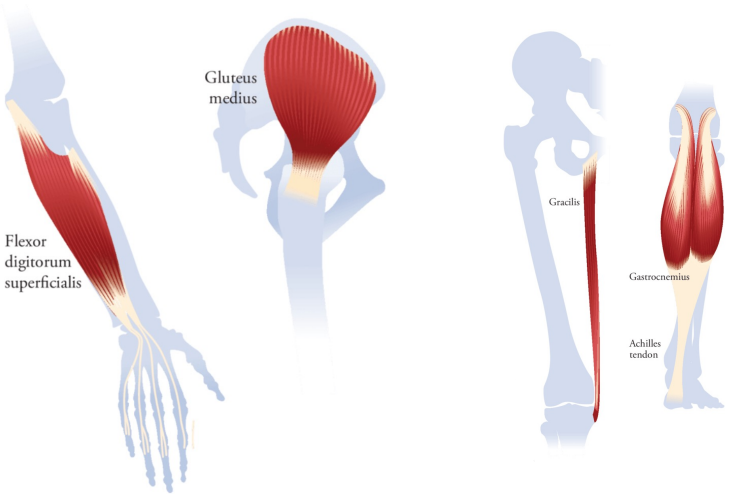


- Can only pull
- Forces applied to skeleton via tendons
- Very resilient tissue
- Performance determined by:
  - Excitability/Irritability: ability to response to stimulation
  - Contractility: ability to generation tension and shorten (shorten on average 57% resting length). *Unique to muscle tissue*
  - Extensibility: ability to lengthen or stretch
  - Elasticity: ability to return to resting length after stretch.

} *Connective tissues/ passive properties*

25

Attach to bone and span joints



Architecture, attachments and geometry dictate function

Uchida & Delp 2020

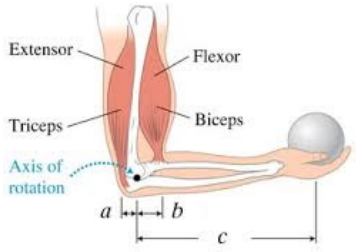
26

Functions

- Produce movement
- Maintain postures and positions
- Stabilize joints
- Others:
  - Support and protect internal organs
  - Control pressure in internal cavities
  - Produce heat, maintain body temperature
  - Control exits and entrances

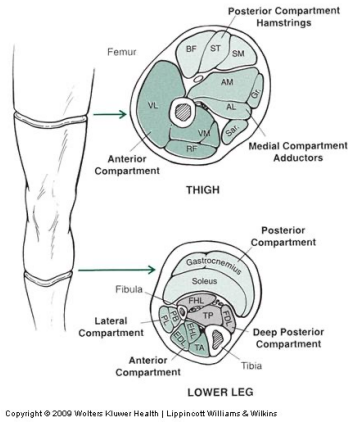
27

Described in functional groups



28

Groups of Muscles: synergistic muscles

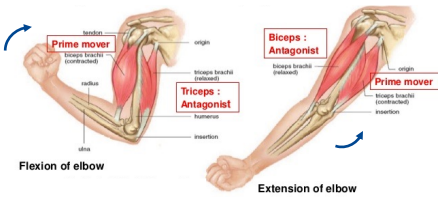


Muscles typically act in groups, not individually

29

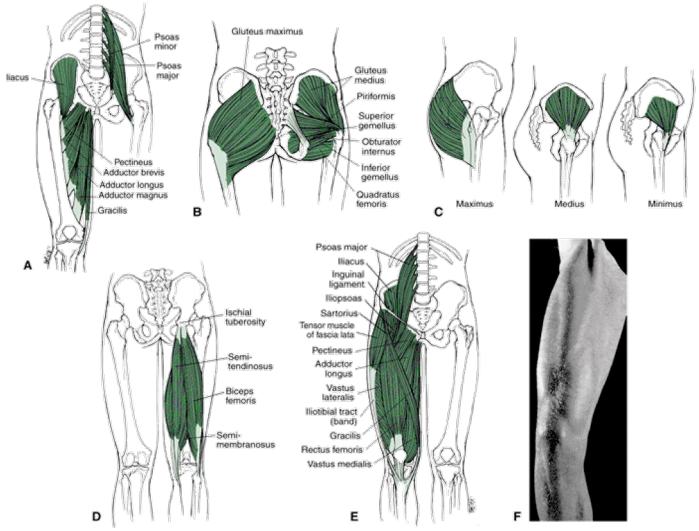
Agonist/Prime mover and Antagonist

- **Specific** to the desired motion; prime mover(s) or **agonist**(s) performs intended motion
- Any muscle that opposes the action of the prime mover is an **antagonist**
- E.g. biceps brachii opposes the action of the triceps brachii to extend the elbow joint. The reverse is true to flex the elbow joint.
- Net moment can be **zero**
  - if there is **no contraction** in agonists and antagonists
  - in **full co-contraction** – if e.g. flexor moment and extensor moment are in balance



30

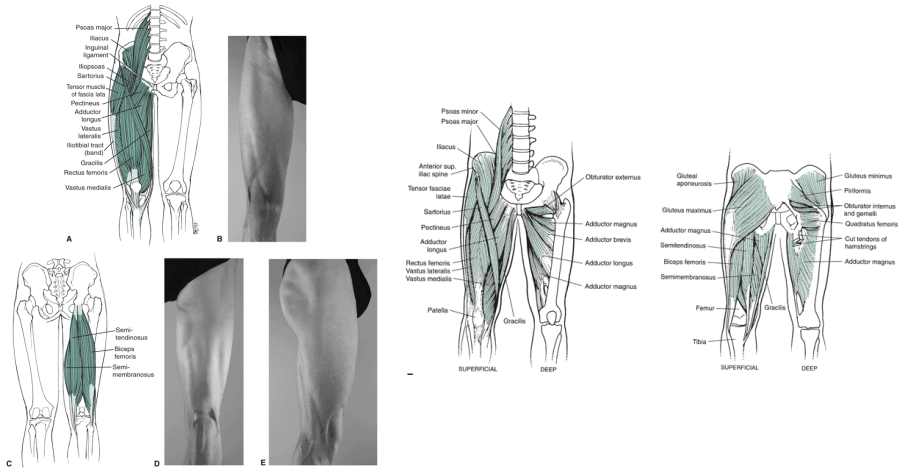
Major hip flexors, extensors, abductors and adductors



Hamill & Knutzen 2009

31

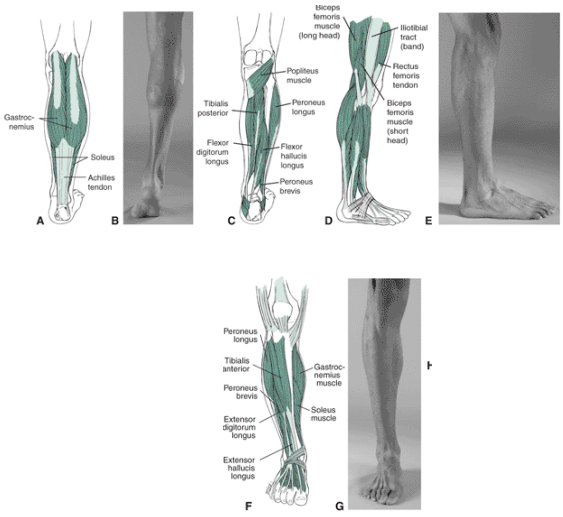
Major knee flexors, extensors



Hamill & Knutzen 2009

32

Major ankle plantarflexors, dorsiflexors, pronators and supinators



Hamill & Knutzen 2009

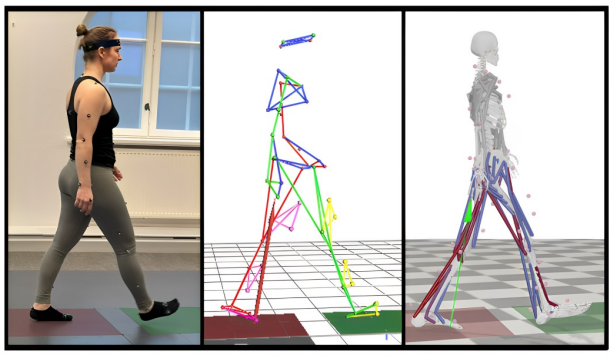
33

Muscle

- Types:
  - striated (striped)
    - Cardiac (not under voluntary control)
    - Skeletal (under direct voluntary control)
  - non-striated (smooth)
    - Not under voluntary control
    - Ex. Veins, arteries, bladder, iris of eye

34

Tomorrow: Motion analysis



35

About Thursday 30 October

Compulsory to attend one of the three sessions! You can sign up on Canvas calendar

Be there at least 5 minutes before

9.15 – 10.00
10.15 – 11.00
11.15 – 12.00

- Promobilia MoveAbility Lab, Osquars Backe 18, 4th floor. Just across from library entrance.

36